

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

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6 September 2026

Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. The physical core is a noisy mean-field phase model whose coupling varies slowly and crosses a synchronization threshold: if recovery of coherence after sleep is gated by such a crossing, it is not free to take an arbitrary time course: for as long as the phase distribution tracks the stationary branch it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. That tracking condition is part of the prediction rather than a caveat on it, and the simulations measure what it costs. Ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity that descends a squared-drift objective preserves coherence while moving away from the environmental structure it is expected to learn, below the level of a shuffled-label comparison, which is evidence against the common

intuition that dissipation minimization implies representational learning. A Lean 4 formalization composes finite information capacity, Landauer’s bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem and isolates the eight hypotheses that composition consumes; a common toy model satisfies all eight. We state and defend, as a philosophical commitment rather than a theorem, the identification of the glued global section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. The decisive empirical requirements are independent calibration of effective cortical coupling and an awakening experiment that separates slow coupling change from alternative recovery mechanisms.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the collective state such accounts require remains an empirical question.

Such an account has three requirements, and they order what follows. Coupling must be strong enough to sustain coherent collective activity; the local descriptions that activity carries must be compatible enough to glue into a single one; and that glued state must support an accurate model of

15 itself. The first requirement is physical and testable with current methods,
16 and most of this article is about testing it. The second and third are where
17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery
30 hypothesis itself: a stationary coherent branch with square-root onset, a von
31 Mises concentration–coherence relation, and an experimental protocol with
32 failure conditions declared in advance (Sections 2 and 3). The second is a
33 constraint on how such data may be analyzed. The apparent onset expo-
34 nent measured from a finite ramp is a function of ramp speed across the
35 sweep of Section 4, so an exponent fitted to a recovery trace measures the
36 ramp at least as much as the mechanism—and that holds for any analysis
37 reading a critical exponent off a driven transition, not only for ours. The

38 third is a negative result about plasticity: a coupling update descending
39 a squared-drift objective preserves coherence while moving *away* from the
40 environmental structure it is expected to learn, below a shuffled-label com-
41 parison. Dissipation minimization does not imply representational learning.
42 The fourth is a machine-checked composition (Section 5) in which finite in-
43 formation capacity, Landauer’s bound, a predictive dissipation bound, the
44 noisy Kuramoto transition, sheaf gluing and Banach’s fixed-point theorem
45 yield one conditional theorem, with the eight hypotheses that composition
46 consumes isolated and named; a common toy model satisfies all eight, so
47 their conjunction is not empty.

48 A prose table of assumptions would list the same eight items, so the fourth
49 contribution needs a word about method. What the Lean development adds
50 is that it makes the list non-negotiable. The characteristic failure in an
51 argument of this shape is not a false step but a silent change of subject:
52 a theorem about a stationary density read as a claim about a trajectory,
53 an inequality about a register read as a claim about learning, agreement
54 of phases read as compatibility of representations. Type-checking forbids
55 each of those substitutions, and an assumption needed to make a step go
56 through appears as a hypothesis in the term rather than as a word in a
57 sentence. Formalization does not make the assumptions true; it makes them
58 impossible to lose.

59 The physical model and the formal composition belong in one article be-
60 cause they are two halves of one commitment. The coupling-gated prediction

61 is worth testing under this framework, rather than as one more synchroniza-
62 tion model, only because the coherent state it produces is claimed to do
63 further work: to support a globally consistent representation and a reflexive
64 fixed point that we identify with phenomenal unity and a minimal self.

65 The distinction that does that work is between phases agreeing and de-
66 scriptions fitting together, and it is concrete. Take two cortical regions whose
67 receptive territories overlap, each carrying a probability distribution over
68 what occupies the shared part. Their oscillators can be locked to the same
69 phase at every instant while one region's distribution on the overlap places
70 an object left of fixation and the other places it right. The phases agree;
71 the descriptions do not, and no single distribution over the shared terri-
72 tory restricts to both. A theory that identifies unity with synchrony cannot
73 mark that difference, and so cannot separate a brain whose regions oscillate
74 together from one whose regions are about the same world. The sheaf con-
75 dition marks exactly it: local states glue to one global section when their
76 restrictions to shared regions coincide, and otherwise not at all. Coherence
77 enters as the physical condition under which compatibility can be main-
78 tained, not as a substitute for it, and the self-prediction map then asks the
79 further question of whether the glued state contains an accurate model of
80 itself. That identification is a philosophical commitment rather than a theo-
81 rem, and Section 6 states it, argues for it, and says what would count against
82 it. Its separation from the physical tests is what makes both assessable: the
83 cortical coupling hypothesis and the recovery prediction can fail while the

84 mathematical composition stands, and the composition can stand while the
85 identification is wrong. The established ingredients (Landauer, 1961; Still
86 et al., 2012; Sakaguchi, 1988; Banach, 1922) therefore appear here for their
87 biological composition, not for any new fundamental physical law.

88 **2 A coupling-gated model of cortical coher-** 89 **ence**

90 **2.1 Dynamics and stationary solutions**

91 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

92 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
93 K_{ij} has inverse-time units, with angles treated as dimensionless. The uni-
94 form mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and
95 phase diffusion are separate parameters; a frequency-spread width cannot be
96 substituted for D when applying the identical-frequency stationary theory.

97 For identical frequencies in their rotating frame, the assumed stationary
98 density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

99 The I_n are modified Bessel functions, and ψ is the collective phase. Lean
 100 verifies normalization, the density's circular order parameter, and the self-
 101 consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent
 102 order parameter is zero. At $K > K_c$, exactly one positive solution coex-
 103 ists with zero; the positive branch is continuous at threshold and strictly
 104 increasing with K .

105 The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

106 This is a result about the stationary solution branch. Of the coherent branch
 107 we prove existence, uniqueness, continuous emergence at threshold and strict
 108 growth in K , but not that it is dynamically selected. The dynamical mean-
 109 field limit—propagation of chaos—is the missing link between the finite sys-
 110 tem and the stationary description (Sznitman, 1991), and no theorem in this
 111 development closes it. Exchange symmetry alone does not justify factoriza-
 112 tion of the joint law, particularly for an ensemble with an unpinned collective
 113 phase.

114 A spatial version replaces the finite sum by an integral against a kernel
 115 $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-
 116 field strength is a kernel average rather than a quantity that grows with the
 117 arbitrary measure of the substrate. Reduction of a heterogeneous spatial
 118 kernel to a scalar K is an additional modelling step, and the threshold in

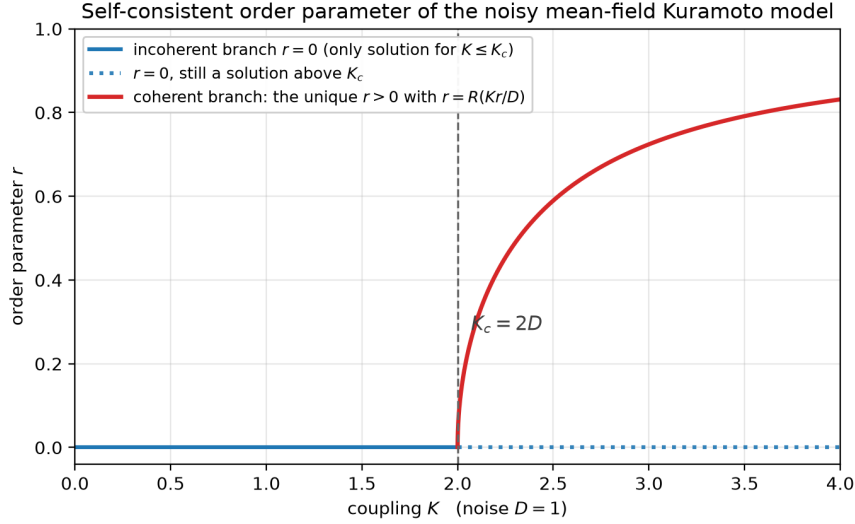


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

Eq. (2) does not automatically apply to every spatial or signed network.

2.2 Physical identification and calibration

Regular mesh refinement gives convergence of a discrete coupling-energy quadrature to its continuum integral. This is scalar energy convergence, not convergence of trajectories and not a construction of a physical kernel. The model posits a continuum kernel and separately identifies its cortical realization with endogenous electromagnetic coupling. The formal predicate requires a jointly continuous kernel, a normalized substrate, a mean-field

127 strength equal to its spatial average, and no single site providing more than
 128 half of that mean-field coupling. Toy witnesses satisfy these conditions; none
 129 is a model calibrated to cortex.

130 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
 131 2010; Anastassiou & Koch, 2015), and field-based models have been proposed
 132 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
 133 Fadden, 2020). These observations motivate an effective interaction without
 134 establishing its magnitude relative to phase diffusion, its spatial averaging
 135 rule, or its causal role in experience; changes in phase organization under
 136 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s
 137 coupling threshold.

138 The physical-unit conversion is the sharpest open limitation of the frame-
 139 work, and we state it as such rather than working around it. An estimate
 140 proportional to $N\Delta tf$, using population size, a spike-time shift and oscil-
 141 lation frequency, is dimensionless. It cannot be compared directly with an
 142 inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

143 requires a rate calibration γ , where s is the time shift per unit field and E
 144 the field amplitude. The saved field-strength calculations are conditional on
 145 this calibration and on the chosen spatial aggregation; they do not indepen-
 146 dently establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction kernel

147 and phase diffusion from the same physical system is necessary before the
 148 threshold becomes a quantitative biological claim, and Section 3.2 makes
 149 that measurement the first item of the proposed experiment rather than an
 150 assumption of it.

151 **3 Recovery after awakening: a conditional** 152 **prediction**

153 Sleep-associated changes in extracellular space and clearance motivate a pos-
 154 sible slow control of coupling (Xie et al., 2013). We hypothesize that recovery
 155 after awakening includes a change in effective coupling $K(t)$, mediated by
 156 extracellular geometry, on a slower timescale than phase dynamics. Neither
 157 the monotonic relation between geometry and coupling nor its role in hu-
 158 man sleep inertia follows from the cited physiological observation; the model
 159 requires both to be measured.

160 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 161 stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

162 The ideal stationary description has a subthreshold interval followed by con-
 163 tinuous onset. Saturation requires a specified subsequent coupling trajectory;
 164 monotonic coupling alone does not determine a logistic or any unique sig-

165 moid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently
 166 known. When coupling is inferred from the same coherence trace, this is not
 167 a parameter-free prediction of latency.

168 Critical slowing down limits the adiabatic prediction of Eq. (5) near
 169 threshold, and a finite population carries nonzero fluctuating order below
 170 it. An observable onset therefore requires a predeclared threshold above the
 171 finite-size floor, and a finite-rate trajectory need not show an infinite-slope
 172 cusp. Section 4.1 quantifies the limitation.

173 **3.1 Distributional and temporal tests**

174 The stationary model makes a second prediction: phase concentration and
 175 coherence obey $r = I_1(a)/I_0(a)$. This relation has no fitted shape parameter,
 176 but its applicability depends on the stationary-density assumption and the
 177 observation model. It is common to the von Mises family and therefore
 178 cannot by itself identify ephaptic coupling or distinguish this framework from
 179 every alternative synchronization model. Within the same ansatz, $a/r =$
 180 K/D for $r > 0$; the limiting threshold ratio is a consequence of the density
 181 relation, not an independent test of it, and ratios near the finite-size floor
 182 are especially unstable.

183 The estimators must be independent, which is the constraint most easily
 184 violated in practice. Estimating a by inverting the Bessel ratio from r would
 185 enforce the relation being tested, and linear statistics on wrapped phases
 186 estimate neither quantity.

187 **3.2 A protocol for the awakening experiment**

188 The requirements above define a design rather than a single measurement,
189 and the design is where the content of the hypothesis lives. An adequate test
190 needs all six of the following.

- 191 **1. Coupling and diffusion in the same units.** Estimate K and D
192 from the same cortical regime, in inverse-time units, separating phase
193 diffusion from quenched frequency spread and from observation noise.
194 Without this the threshold $K > 2D$ is not an evaluable statement
195 about cortex.
- 196 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or
197 an astrocytic signal is an indicator of extracellular geometry, not a mea-
198 surement of K . Its mapping to effective coupling must be established
199 separately, with its own uncertainty.
- 200 **3. Phase estimation at adequate spatial resolution.** At least 100
201 spatial samples per estimate. Estimate r by the circular resultant
202 length and a by an unweighted regression of log-density on the co-
203 sine of phase relative to the circular mean. With fewer sites, pooling
204 over time changes the estimand to a spatiotemporal distribution and
205 must be stated as such.
- 206 **4. An independent recovery endpoint.** Behavioral recovery measured
207 separately from the phase trace, so that latency is not defined by the

208 signal whose latency is at issue.

209 **5. A predeclared onset criterion.** A coherence threshold fixed above
210 the finite-size floor before unblinding, chosen in the knowledge that a
211 finite ramp smears the cusp.

212 **6. Model comparison under matched observation.** Coupling-gated
213 recovery compared against exponential, flexible sigmoidal and other
214 slow-recovery models under identical temporal filtering and observation
215 noise. Latency alone cannot distinguish the mechanism, because many
216 processes create delays.

217 The design fails informatively in three ways: failure of the stationary
218 phase-density relation over a sufficiently broad concentration range; absence
219 of the proposed coupling–geometry association; or a recovery trace better
220 explained by an alternative after finite-rate and observation effects are ac-
221 counted for. Two apparent failures are not failures. A linear apparent onset
222 is evidence against the adiabatic approximation only where that approxima-
223 tion is independently justified, and it does not falsify the field hypothesis
224 under arbitrary ramp speeds. The absence of a fold in the stationary equa-
225 tion excludes stationary coexistence of two positive branches in this model,
226 but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

227 4 Numerical controls

228 The simulations characterize the finite-size, finite-rate regime that the sta-
229 tionary theory does not cover. They are reported here, ahead of the formal
230 composition, because they bear directly on whether the prediction of Sec-
231 tion 3 is measurable; none of them discharges a hypothesis of the composition
232 in Section 5. Each control uses explicit seeds and compact saved summaries,
233 and the numerical values below are generated from those artifacts. The
234 supplement gives parameters, estimators, failure criteria and additional pan-
235 els. Identical-frequency controls use phase diffusion rather than a quenched
236 frequency-spread surrogate, and order plots display the $1/\sqrt{N}$ reference.

237 4.1 Ramp speed sets the apparent onset exponent

238 The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at
239 four coupling speeds. Escape is the first of three consecutive saved samples at
240 $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window.
241 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,
242 with a fitted delay exponent 0.443. The apparent onset exponent changes
243 from 0.964 at the fastest speed to 0.417 at the slowest.

244 The result cuts against the prediction of Section 3 as much as against
245 any other reading of a driven transition. The exponent recovered by fitting
246 such a transition is a joint function of the mechanism and the drive rate. Re-
247 porting an onset exponent from a recovery trace therefore says little about

248 the underlying transition unless the ramp rate is independently known and
 249 the finite-size floor is declared; the stationary square-root law cannot be im-
 250 posed on an arbitrary recovery trace, and its absence is not by itself evidence
 251 against a coupling-gated mechanism. The specified deviation-based critical
 252 ramp speed is not bracketed by the sweep.

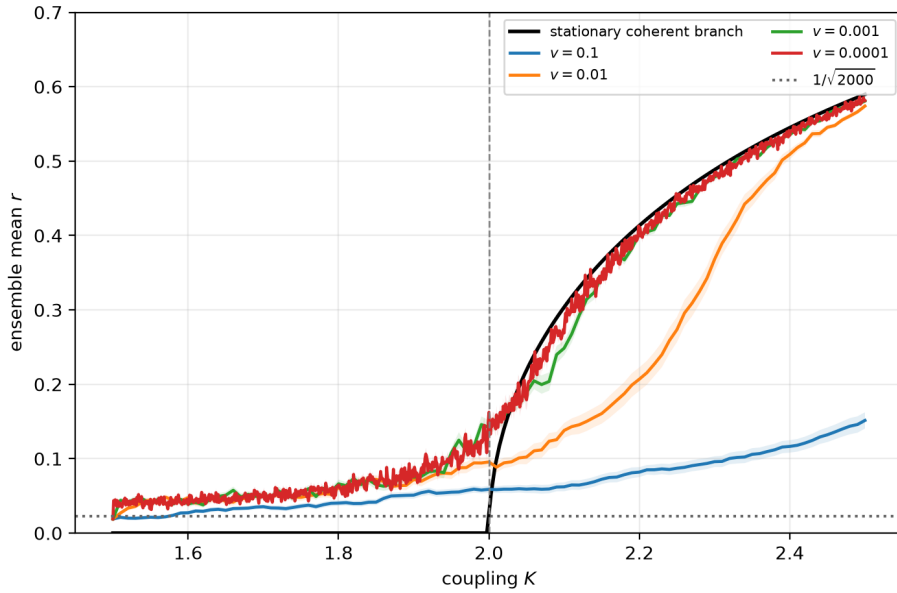


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

253 The dynamical-selection control shows escape toward a coherent branch
 254 above threshold, but retains quantitative finite-size discrepancies. At $K =$
 255 2.8, steady mean order is 0.67735 against the static value 0.68270. The

256 fitted growth-rate slope is 0.43240, compared with the theoretical slope $1/2$.
 257 The propagation-of-chaos diagnostics show decreasing subcritical dependence
 258 with increasing population size. In the supercritical unpinned ensemble,
 259 however, pair correlation remains substantial because realizations share a
 260 random collective orientation. This negative unconditional result does not
 261 settle a conditional mean-field limit after removal of that orientation.

262 4.2 Spatial coupling, frustration and calibration

263 A periodic two-dimensional sheet with row-normalized exponential coupling
 264 remains coherent in the sampled empirical decay-length band under the de-
 265 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
 266 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-
 267 eter regime; fixing total coupling while changing range tests fragmentation,
 268 not whether cortex supplies that total coupling.

269 A row-balanced Dale-law network shows that row balance alone does not
 270 guarantee a floor-level incoherent baseline. In a separately specified weak-
 271 synapse regime, a uniform modulatory term produces an operational co-
 272 herence crossing at K_{eff}/D between 1.8994 and 1.9387. The physical field
 273 conversion remains conditional on the rate calibration γ . Neither the spa-
 274 tial control nor the rescue sweep measures cortical coupling or resolves the
 275 unit-conversion limitation.

276 4.3 Squared-drift plasticity does not learn structure

277 A coupled system that descends a dissipation-like objective is widely expected
 278 to come, in doing so, to mirror the structure of its input; the predictive
 279 dissipation bound of Section 5 invites that reading. This control tests the
 280 expectation directly in the present model, and the expectation fails.

281 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

282 It is a quadratic functional of an affine image of the kernel. The formalization
 283 constructs the bounded drift operator A and its adjoint on the relevant L^2
 284 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
 285 decreases this objective at fixed phases. The objective is not generally a
 286 thermodynamic entropy-production rate, which requires probability-current
 287 information. Whether biological couplings follow this flow is an additional
 288 modelling question.

289 The joint control evolves noisy phases with clustered frequencies and
 290 nonzero grand-mean drive, while updating symmetric couplings every 50 fast
 291 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-
 292 tal. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so both
 293 endpoints contribute. Resource constraints and evolving phases remove the
 294 fixed-phase monotonicity guarantee.

295 Across three seeds, second-half order lies between 0.9811 and 0.9815, and

296 matched frozen-coupling controls maintain similar coherence. The distance
297 to the true cluster template increases in every adaptive run, with worse
298 changes than the shuffled-label comparison (Fig. 3). The template is a co-
299 variance of shared latent cluster signals, not an input to the update or a
300 measured covariance of the constant frequencies.

301 The objective descends, coherence is maintained, and the coupling matrix
302 moves away from the generating structure—below the level of a random
303 relabelling. Coherence is therefore not a proxy for representational alignment
304 in this regime, and a plateau in the objective establishes neither stationarity
305 of the adaptive process nor a thermodynamic entropy-production rate.

306 4.4 The estimation protocol demonstrated on public 307 EEG

308 Item 3 of Section 3.2 specifies estimators whose behavior on real recordings
309 is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa
310 et al., 2025). The purpose is to show what the protocol produces and where
311 it is fragile, not to test the awakening hypothesis, for which these data are
312 the wrong design.

313 Two fragilities appear immediately. The implemented estimator pools 62
314 channels over a 100-ms window, so it estimates a spatiotemporal distribu-
315 tion rather than an instantaneous spatial one—a departure from the protocol
316 forced by channel count, and one that any scalp implementation will face.

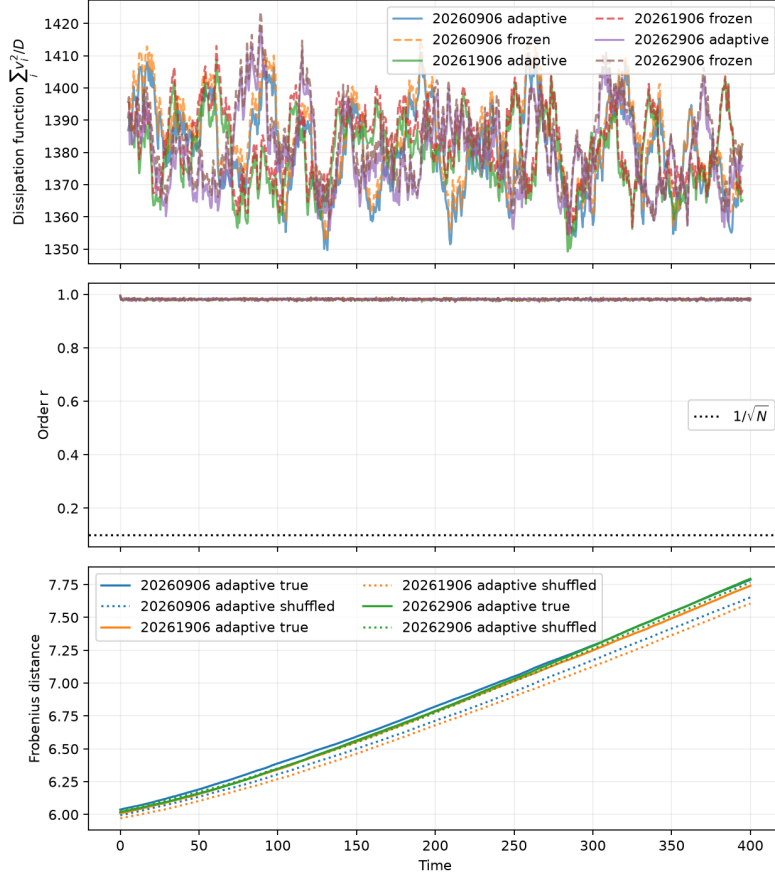


Figure 3: Joint fast/slow dynamics with matched frozen-coupling controls. Top: smoothed squared-drift dissipation, with frozen controls dashed. Middle: order and the finite-size reference. Bottom: distances to true cluster templates (solid) and node-label shuffles (dotted). Coherence persists, but structural distances increase. The result limits the proposed connection between dissipation minimization and environmental structure.

317 Bipolar and common-reference montages give different residuals, making ref-
318 erencing a material analysis choice rather than a preprocessing detail.

319 The bipolar results are compatible with the Bessel relation over the ob-
320 served concentration range. That range is too narrow to distinguish the
321 curve from its linear approximation at the observed scatter, so the check is
322 uninformative about the relation itself. An informative test needs a broader
323 concentration range and explicit treatment of source mixing, finite samples
324 and temporal pooling; intracranial or high-density recordings across a gen-
325 uine state transition are the natural setting.

326 5 Thermodynamic constraints and the for- 327 mal composition

328 The synchronization model of Section 2 is one link of a longer conditional
329 construction, whose nine nodes are:

330 **n1** a finite phase space has bounded information capacity; **n2** a
331 broken symmetry forces a boundary, where the field leaves its vac-
332 uum manifold; **n3** the boundary erases, and erasure costs heat;
333 **n4** that dissipation bounds nonpredictive memory; **n5** discrete
334 couplings coarse-grain to a continuum limit; **n6** a specified phys-
335 ical field realizes that limiting coupling; **n7** above threshold the
336 field carries a unique coherent order parameter; **n8** compatible
337 local states glue to a unique global section; **n9** that section is the
338 unique fixed point of a self-prediction map.

339 Node-level results are established or cited; what the composition consumes

are the eight implications between them. Table 1 lists these, classified by their role in the composition rather than by their plausibility. In particular, finite capacity does not imply a symmetry-breaking boundary: n_1 and n_2 are independent roots, even though the composition can accept an implication between them as an argument.

5.1 Capacity, boundaries and predictive memory

For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with equality only for the uniform distribution. A stream with more possible histories than $|X|$ cannot be encoded injectively in that register. Separately, a continuous field on a connected substrate that reaches distinct components of a disconnected vacuum manifold must leave that manifold somewhere, the simplest topological-defect construction (Kibble, 1976). Relating this locus to a particular finite information-processing register is the additional hypothesis E12 carries.

A non-surjective update on a finite register is non-injective. Under the specified system–bath model and thermal assumptions, erasure carries a Landauer heat cost (Landauer, 1961). Finite capacity does not imply that every update erases, and logical many-to-one processing must be distinguished from its physical implementation (Norton, 2005; Shenker, 2000). The formalization records the physical heat identity as an obligation of each model and supplies an explicit one-bit eraser satisfying it.

For a system state X_t driven by an environmental signal S_t , the predictive

Edge	Status	Required connection
E12 capacity $\rightarrow$ boundary	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.
E23 boundary $\rightarrow$ erasure	Formalization gap	The boundary is associated with a finite register whose update leaves states unreachable.
E34 erasure $\rightarrow$ prediction	Formalization gap	Erasures supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
E45 prediction $\rightarrow$ continuum	Modelling assumption	A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
E56 continuum $\rightarrow$ field	Physical commitment	The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.
E67 field $\rightarrow$ coherence	Physical commitment	That field operates above the synchronization threshold.
E78 coherence $\rightarrow$ unity	Formalization gap	Coherence supplies a cover with coupling at least the named mean-field value, reached by relaxation and carrying compatible local states.
E89 unity $\rightarrow$ self	Modelling assumption	The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

362 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

363 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
364 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains
365 nonpredictive memory at a given dissipation budget. It does not specify
366 a learning rule, maximize prediction, or require the internal representation
367 to reproduce environmental geometry—the point the plasticity control of
368 Section 4 makes concretely. Connecting this budget to Landauer heat ad-
369 ditionally requires nonpredictive information to be bounded by the entropy
370 erased by the register. The formal bridge derives Eq. (7) from Landauer’s
371 bound under that identification; it does not derive the identification.

372 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
373 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-
374 tropy, not a general identification of relative entropy with physical entropy
375 production. These distinctions prevent an informational inequality from be-
376 ing read as a derivation of environmental learning, Bayesian inference, or
377 semantic content.

378 5.2 Compatible states and reflexive fixed points

379 A sheaf formalizes local descriptions and their restriction to smaller regions.
380 If local sections agree on every overlap, the sheaf condition gives a unique

381 global section. In the probability-sheaf construction, this is consistency of
382 local density information, and it is not implied by agreement of oscillator
383 phases: the development includes a counterexample with equal phases but
384 incompatible densities on an overlap. Section 6 rests on this distinction,
385 and the model excludes such configurations only through its compatibility
386 hypothesis.

387 A thermodynamic cover couples this gluing construction to a phase con-
388 figuration at a potential minimum. For the deterministic positive-coupling
389 model, convergence to that minimum is proved under an explicit initial basin
390 condition; other equilibria prevent an unconditional conclusion. The station-
391 ary noisy order parameter, the deterministic relaxation, and the cover must
392 therefore be connected by E78 rather than identified without a hypothesis.
393 Neither field continuity nor synchronization alone guarantees a globally com-
394 patible representation.

395 A reflexive extension equips global sections with a complete metric and a
396 predictive map formed by restricting a state to a local encoding region and
397 reading that encoding out. If the map contracts and fixes the section supplied
398 by the cover, Banach’s theorem gives uniqueness of that same fixed point.
399 The contraction rate is assumed, and the equality between the glued section
400 and the fixed point is part of E89. A coherent order parameter supplies the
401 supercritical inequality required by the stated contraction bound; it does not
402 construct the map or prove that cortical dynamics implements it.

403 The theorem `chain` concludes `UnifiedSelf` from these hypotheses. A

404 common toy witness satisfies all eight, so their conjunction is not empty.
405 The project declares no additional Lean axioms beyond the foundational
406 axioms used by Lean’s libraries; physical content resides in hypotheses and
407 model fields. This checks the conditional mathematical composition, not the
408 empirical truth of its premises. Detailed constructions, scope statements and
409 counterexamples are retained in the supplement and source files.

410 **6 What the construction commits to about** 411 **consciousness**

412 The composition terminates in a mathematical object, and we take a posi-
413 tion on what that object is. The commitment has two clauses. *Unity*: the
414 unique global section glued from compatible local states is the formal corre-
415 late of the unity of a conscious episode—the fact that a subject undergoes
416 one experience rather than a collection of co-occurring local ones. *Selfhood*:
417 the fixed point of the self-prediction map on that same section is the formal
418 correlate of a minimal self, a state that is what it takes itself to be. Neither
419 clause is a theorem, and no simulation here supports either. We argue for
420 them, then say what would count against them.

421 Three features of the construction motivate the identification, and each
422 distinguishes it from a bare synchrony measure.

423 First, compatibility rather than agreement. Oscillator phases can agree
424 everywhere while the local probabilistic descriptions they carry remain in-

425 compatible on their overlaps, and Section 5.2 exhibits that configuration
426 inside the formalization rather than only as an illustration. If the unity of
427 experience is a matter of local contents being about one world rather than
428 merely co-timed, then compatibility, not coherence, is the property to look
429 for—and coherence enters as the physical condition under which compatibil-
430 ity can be maintained, which is what E78 asserts.

431 Second, uniqueness follows from the construction rather than from a sep-
432 arate selection principle. Accounts of conscious unity need something that
433 picks one experience from among the candidates a system could support;
434 integrated information theory postulates exclusion for this purpose (Tononi
435 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover
436 whose local sections agree, the sheaf condition gives exactly one global sec-
437 tion, and given a contracting self-prediction map, Banach’s theorem gives
438 exactly one fixed point. The scope of that claim needs stating exactly, be-
439 cause the stronger reading is available and is not ours. Uniqueness is relative
440 to a cover and a map, and choosing them—which regions, which overlaps,
441 which encoding region, and so where the subject ends—is not supplied by ei-
442 ther theorem. It is what E78 and E89 carry. What the construction removes
443 is the need for an additional principle to select among the experiences a fixed
444 cover and map would otherwise admit; it does not remove the need to say
445 which cover and map a given subject has. That is a structural argument,
446 not evidence.

447 Third, the reflexive clause makes the self a condition rather than an addi-

tion. The predictive map restricts a global state to a region encoding it and reads that encoding back out, so a fixed point is a state whose self-model is accurate about the state it models. That condition is available only because the state modelled is a single glued object: a system whose local descriptions do not cohere has no such state for a self-model to be about. Unity and selfhood are therefore one construction read at two levels, which is why `UnifiedSelf` requires the fixed point to be the glued section itself rather than an unrelated section.

The costs belong with the argument. The commitment does not address why any physical condition is accompanied by experience; it is a proposal about which physical state is the conscious one, not about why states of that kind feel like anything. It is therefore open to the objection that a system could satisfy every condition—compatible local states, a supercritical field, an accurate self-model—while nothing is experienced. We have no reply to that objection and claim none. The identification is also multiply realizable as stated: nothing in the sheaf or fixed-point conditions is specific to cortex or to electromagnetic coupling, so the substrate claim is carried entirely by E56 and E67 and is empirical. No conclusion about artificial systems follows from the formal results.

What could count against the commitment is dissociation. It predicts that the two clauses and the presence of a unified conscious episode go together, so a state in which local cortical descriptions are demonstrably compatible and the field supercritical, while the subject reports no unified experience,

471 would tell against it; so would unified reported experience where compati-
472 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
473 states are the places to look, because they separate coherence from report
474 in ways ordinary waking does not. Such tests need an operational measure
475 of overlap compatibility on neural data, which this article does not supply;
476 constructing one is the main obstacle to making the commitment empirically
477 live rather than merely stated.

478 The physical tests of Sections 3 and 4 are independent of all of this. They
479 concern coupling, coherence and recovery, and can fail while the identification
480 stands, or succeed while it is wrong.

481 **7 Discussion**

482 The framework connects thermodynamic information constraints, collective
483 phase organization and compatible local representations, conditionally and
484 with its assumptions named. Its biological content is a coupling-gated re-
485 covery hypothesis with defined observables, a stated protocol, and failure
486 conditions declared in advance. Its formal content is a composition whose
487 eight gaps are exhibited rather than argued past, and which can be inhabited
488 by a single model. Its interpretive content is the identification of Section 6,
489 offered as a commitment with reasons.

490 The simulations are useful because they restrict the hypothesis rather
491 than support it. Finite ramps smear the stationary onset to the point where

492 a fitted onset exponent chiefly reports the ramp; an unpinned collective
493 phase makes independence diagnostics ambiguous in exactly the supercritical
494 regime the model cares about; and squared-drift plasticity maintains coher-
495 ence while degrading structural alignment. The second and third of these
496 are results about method and about a common intuition, and they transfer
497 beyond this framework.

498 The proposal differs in emphasis from accounts based on integrated in-
499 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
500 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
501 candidate interaction medium and a mathematical compatibility condition,
502 and, relative to a given cover and self-prediction map, it derives the unique-
503 ness those accounts postulate. It does not derive the exclusion principle of
504 integrated information, identify every broadcast with a field transition, or
505 derive a generative model from a dissipation bound. Continuous fields can
506 carry incompatible local descriptions and discrete models can admit sheaf
507 constructions, so continuity alone does not settle the combination problem;
508 compatibility is what this account offers in its place.

509 Two formal results bear on the substrate question, and both admit a
510 stronger reading than they support. At matched coupling resource, a fixed
511 wiring support that excludes a better-correlated pair can be suboptimal—a
512 statement about constrained optimization, not about biology. On an atom-
513 less substrate, a kernel supported on finitely many points contributes zero to
514 the chosen continuum coupling integral. That is a property of the measure

515 and the observable, and it cuts against this framework’s own coarse-graining
516 step as much as against anything else. It does not imply that digital hard-
517 ware cannot approximate a field or support consciousness, and no biological-
518 versus-artificial conclusion follows.

519 The empirical priorities follow directly from Section 3.2: estimate effec-
520 tive coupling and diffusion in consistent units, test the proposed relation
521 between coupling and extracellular geometry, and measure recovery against
522 an independently defined behavioral endpoint under matched model compar-
523 ison. A favorable concentration–coherence relation would support a statis-
524 tical description, not identify its physical cause; an unfavorable result in a
525 regime satisfying the stated assumptions would constrain that description.
526 Successful synchronization simulations cannot substitute for measurements
527 distinguishing ephaptic interactions from other coupling mechanisms.

528 The article therefore asks to be judged in three separable parts: a pre-
529 diction that can be tested, a composition that can be checked, and an iden-
530 tification that can be argued about. Keeping them separate is what makes
531 the whole assessable.

532 **Data and code availability**

533 The Lean development, simulation scripts, tests and saved numerical sum-
534 maries are maintained in the manuscript source repository. Publication sim-
535 ulation values are regenerated from saved summaries without rerunning pro-

duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
ds005620 (Bajwa et al., 2025); preprocessing and selection are described
in Supplemental Material. The source repository contains the reproducible
analysis rather than redistributing the raw recordings.

Use of AI tools

AI tools assisted simulation-code development and manuscript revision. Machine checking establishes the stated Lean results under their hypotheses; it does not validate biological interpretations or replace scientific review of generated code and prose.

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